

Credit Card Default (Taiwan)

Machine Learning Project Report

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1 Problem Framing and Data Understanding

1.1 Problem Being Solved

This project solves a supervised machine learning problem: predicting whether a credit card client will default on the next month's payment based on demographic information, repayment history, bill statements, and payment amounts.

1.2 Target Variable

The target variable is `default payment next month`. In simple terms, this means whether a person is expected to fail to make the required credit card payment on time in the next billing cycle.

To make this clear for readers with no finance background:

- A credit card has a *minimum payment* due each month.
- If a customer does not pay that required amount by the due date, this is considered a **default** for that period.
- The model predicts this event one month ahead.

Although the dataset was collected in Taiwan, the meaning of default is not country-specific. It represents a general credit-risk event used in banking worldwide.

Label encoding in the dataset is:

- 0 = non-default
- 1 = default

1.3 Classification Type

This is a **binary classification** task because the target has two classes only: default or non-default.

1.4 Expected Challenges

The main expected challenges in this dataset are:

- **Feature engineering needs:** important predictive signals are spread across monthly billing, payment, and repayment-status columns, so engineered aggregate and ratio features are needed.
- **Missing values handling:** missing or unclear entries can reduce model reliability and must be checked and treated carefully.
- **Feature selection decisions:** the dataset contains many related variables, so we must decide which original and engineered features to keep to reduce redundancy and improve generalization.
- **Class imbalance risk:** defaults are typically fewer than non-defaults, so evaluation should emphasize metrics like precision, recall, and F1-score, not accuracy alone.
- **Data noise and coding inconsistencies:** categorical fields may include uncommon/invalid codes and require controlled recoding before modeling.

2 Data Acquisition & Documentation

2.1 Dataset Source

The dataset used in this project is:

- **Default of Credit Card Clients (UCI Machine Learning Repository)**
- Link: <https://archive.ics.uci.edu/dataset/350/default+of+credit+card+clients>

In implementation, the dataset is fetched programmatically from UCI through `ucimlrepo` (dataset id 350), then converted into a working dataframe for preprocessing and modeling.

2.2 Modifications Made

The following modifications were applied to the raw data:

- **Column-name standardization:** original fields (e.g., `X1..X23`, `Y`, and alternative target naming) were mapped to consistent readable names such as `LIMIT_BAL`, `PAY_0`, and `default payment next month`.
- **Target normalization:** the target was normalized to a one-dimensional series and unified under one label: `default payment next month`.
- **Repayment-status harmonization:** `PAY_*` variables were aligned to a shared category space, then remapped into semantic categories (e.g., `paid_in_full`, `revolving`, `delay_2_months`, `delay_3_plus_months`).
- **Categorical recoding:** `EDUCATION`, `MARRIAGE`, and `SEX` were recoded into cleaner categorical labels.
- **Specific label merge:** `EDUCATION` values 5 and 6 were merged into 4 (others) before final category labeling.
- **Invalid category filtering:** rows with `EDUCATION = 0` or `MARRIAGE = 0` were removed as invalid category codes.
- **Duplicate handling policy:** duplicates were intentionally **kept** in the current notebook version (no deduplication removal step).
- **Feature engineering:** aggregate and behavioral features were added, including totals, trends, ratios, and utilization features (e.g., `TOTAL_BILL_AMT`, `AVG_UTILIZATION`, `MAX_UTILIZATION`).
- **Feature-selection cleanup:** redundant predictors were removed (e.g., `BILL_AMT2-BILL_AMT6`, `UTILIZATION_2-UTILIZATION_6`, and `ID` when present).
- **Train/test split:** data was split using stratified 80/20 train-test partition with fixed random seed 42 to preserve class balance.
- **Scaling transformation:** `StandardScaler` was fit on training numeric columns only, then applied to test data to avoid leakage.

3 Data Cleaning & Preprocessing

3.1 Handling Missing Values

Missing values were audited using a data-quality summary table that reported missing counts and percentages per column. In this workflow, no standalone imputation method (such as mean/median/mode filling) was applied. Instead, data quality control was handled by:

- validating target and schema consistency,
- filtering invalid categorical codes (e.g., `EDUCATION = 0`, `MARRIAGE = 0`), and
- preserving valid records for downstream feature engineering.

3.2 Encoding Categorical Variables

Categorical preprocessing focused on semantic recoding and category harmonization:

- repayment status columns (`PAY_0` to `PAY_6`) were aligned to a shared category space,
- repayment values were remapped into ordered semantic categories (e.g., `paid_in_full`, `revolving`, `delay_3_plus_months`),
- `EDUCATION`, `MARRIAGE`, and `SEX` were recoded to clean categorical labels,
- `EDUCATION` values 5 and 6 were merged into 4 (others) before final labeling.

At this stage, categorical variables were standardized as categorical features; a separate full one-hot encoding pipeline was specified for later modeling stages.

3.3 Scaling Numerical Features (If Needed)

Numerical scaling was applied using `StandardScaler` after train/test split:

- numeric columns were selected from `X_train`,
- scaler was fit only on training data,
- the same fitted scaler was used to transform both training and test sets.

This prevents information leakage from test data into training.

3.4 Handling Class Imbalance (If Needed)

Class distribution was checked during the data audit and EDA phases. The notebook handled imbalance conservatively by using a **stratified** train/test split so both classes keep similar proportions in train and test sets. No oversampling/undersampling method (e.g., SMOTE) was applied in this stage.

4 Exploratory Data Analysis (EDA)

4.1 Target Class Distribution

The first step in our analysis is to check the balance of our target variable.

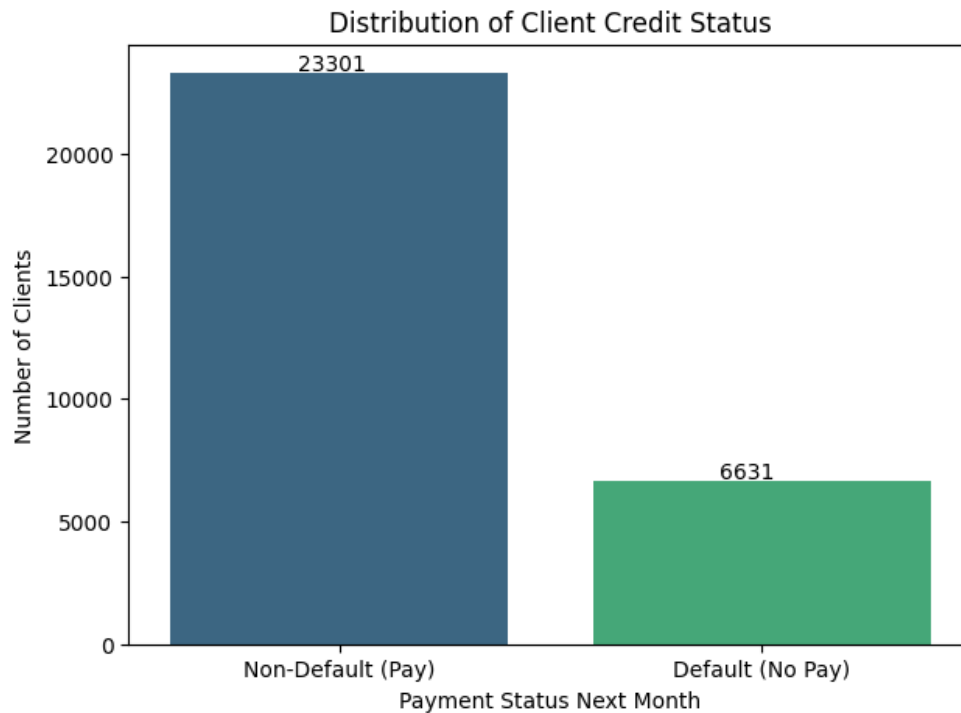


Figure 1: Distribution of Client Credit Status

Interpretation: The dataset exhibits a significant class imbalance, with approximately 22.1% of clients falling into the "Default" category. This imbalance indicates that accuracy alone will not be a reliable metric for model performance. We must prioritize Recall and F1-Score to ensure the model effectively identifies potential defaulters.

4.2 Age Distribution vs. Default Status

We examine how the age of a client correlates with their likelihood of defaulting on payments.

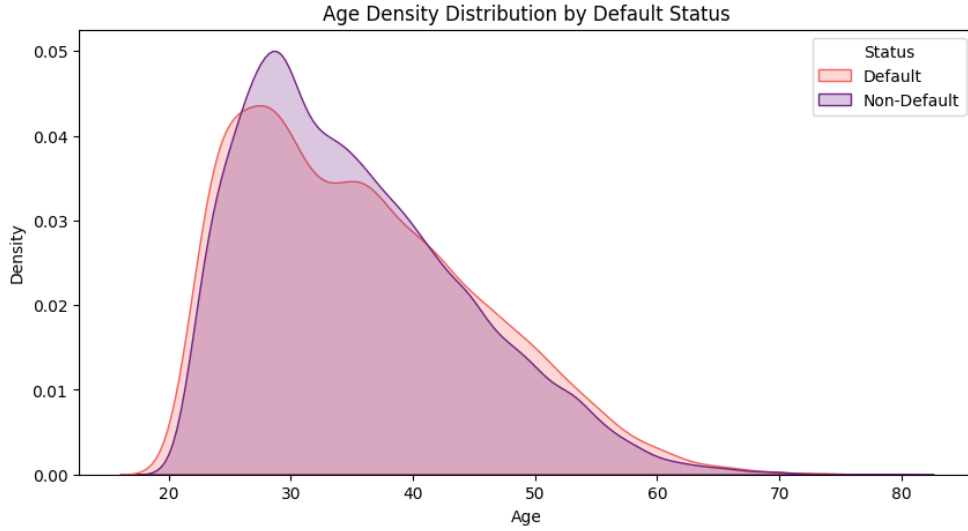


Figure 2: Age Density Distribution by Default Status

Interpretation: The density plot shows that younger clients, specifically those between 20 and 30 years old, have a higher relative density of defaults. As clients age beyond 40, the probability of default tends to stabilize and decrease. This suggests that Age is a relevant demographic feature for credit risk assessment.

4.3 Repayment Behavior (PAY_0) Analysis

This plot explores the relationship between the most recent repayment status and the probability of default.

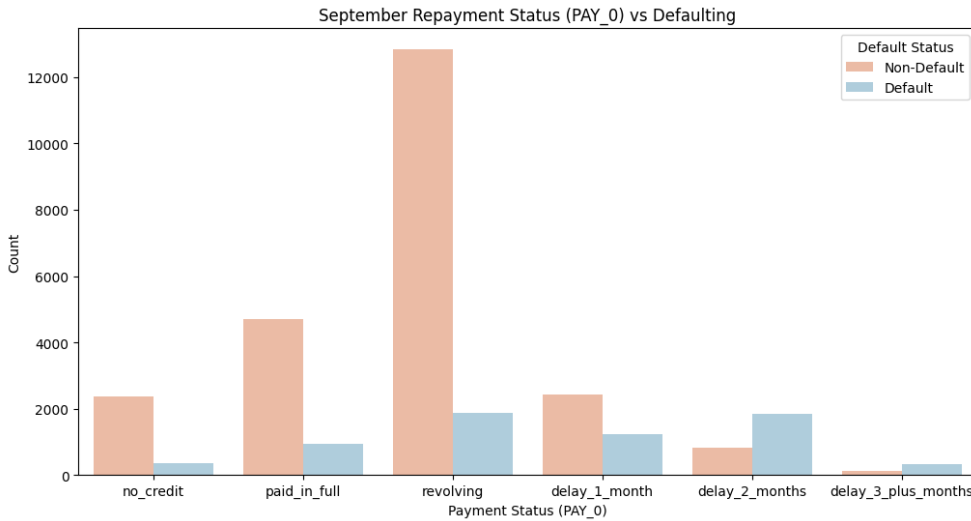


Figure 3: September Repayment Status (PAY_0) vs Defaulting

Interpretation: Clients with a PAY_0 value of 2 or higher (representing a 2-month delay or more) show a much higher ratio of defaults. A status of 0 or -1 (on-time or early payment) strongly correlates with successful repayment next month. Behavioral history, particularly recent delays, appears to be the strongest predictor in this dataset.

4.4 Feature Correlation Heatmap

Finally, we analyze the correlations between numerical features to identify multicollinearity.

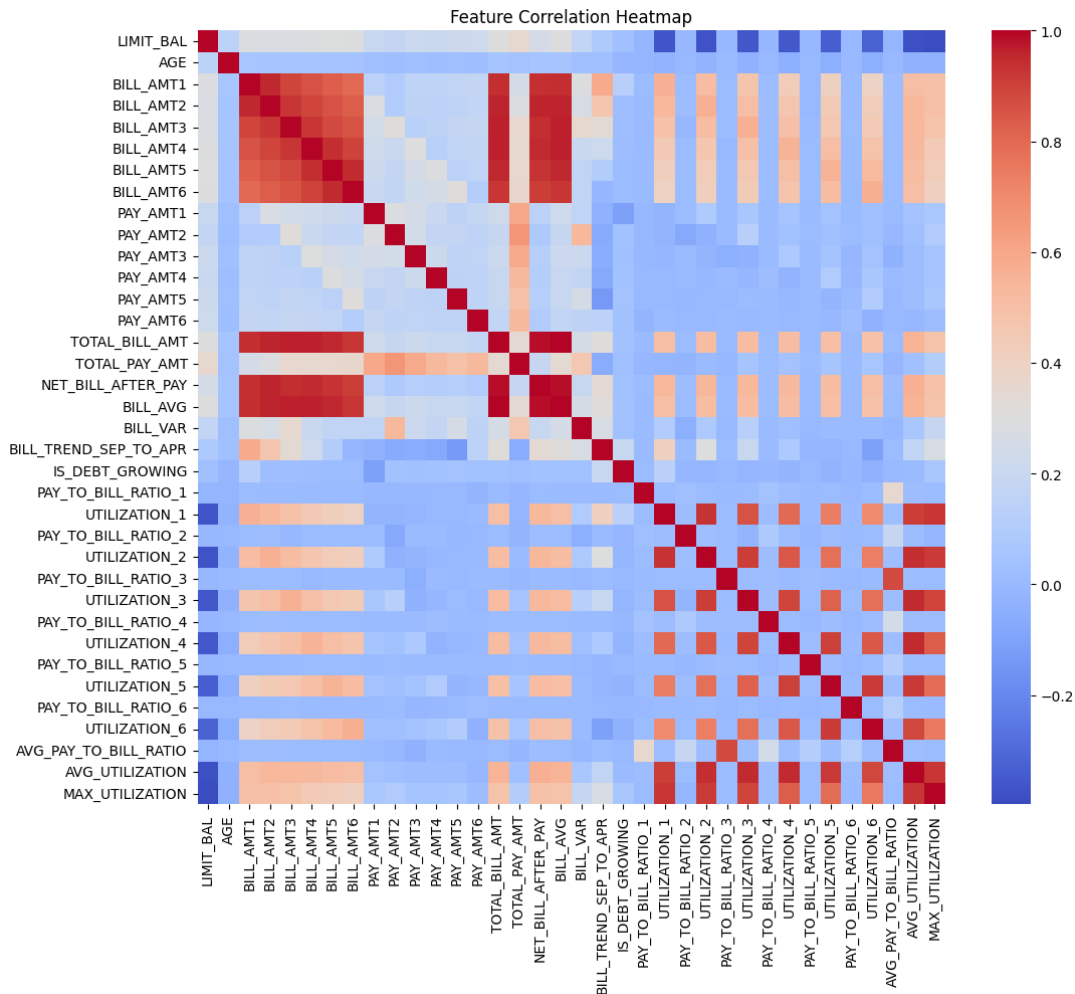


Figure 4: Feature Correlation Heatmap

High positive correlation (approaching 1.0) is observed among the BILL_AMT and UTILIZATION feature groups. This redundancy justifies the removal of several columns to prevent overfitting and improve model stability. Features such as PAY_0 and LIMIT_BAL show distinct correlation patterns with the target, confirming their predictive importance.

5 Modeling — Baseline Classifier (Logistic Regression)

5.1 Model Information

Logistic Regression was selected as the Phase 1 baseline classifier which have:

- **Interpretability:** Coefficients directly quantify each feature's contribution to the predicted default probability. This is valuable in credit risk contexts where explainability is often required.

- **Efficiency:** With $\sim 24,000$ training rows and 35 features, Logistic Regression converges quickly and provides a reliable performance floor for comparison.
- **Class Imbalance Handling:** The `class_weight='balanced'` parameter scales the per-sample loss function inversely with class frequencies, directly addressing the 78/22 imbalance.
- **Regularization:** Default L2 regularization shrinks redundant coefficients and reduces overfitting risk.

5.2 Model Configuration

The model was configured with the following parameters:

Parameter	Value	Justification
<code>solver</code>	<code>lbfgs</code>	Efficient quasi-Newton method; supports L2 penalty natively
<code>max_iter</code>	1000	Ensures convergence without <code>ConvergenceWarning</code> on this dataset
<code>class_weight</code>	<code>balanced</code>	Compensates for the 78/22 non-default/default split
<code>random_state</code>	42	Matches Role 1 and Role 2 seeds for reproducibility

Table 1: Model Configuration Parameters

5.3 Training Pipeline

The model was trained strictly on the training partition. The test set was never accessed during training to prevent data leakage. After the full preprocessing pipeline, the training set contains 23,945 samples and 69 features, while the test set contains 5,987 samples and 69 features. The 69-feature space is the result of the full `ColumnTransformer` pipeline applied by Role 1, including one-hot encoding of all categorical variables (`PAY_0` through `PAY_6`, `EDUCATION`, `MARRIAGE`, `SEX`).

6 Performance Evaluation

6.1 Summary Metrics

The trained model was evaluated on the held-out test set of 5,987 samples. Three primary metrics were computed:

Metric	Value	Interpretation
Accuracy	76.83%	Overall correct predictions — less meaningful due to class imbalance
F1-Score (Default class)	0.5162	Primary metric — harmonic mean of Precision and Recall for defaulters
ROC-AUC	0.7634	Model ranks a random defaulter above a non-defaulter 76% of the time

Table 2: Summary Metrics

	precision	recall	f1-score	support
Non-Default (0)	0.87	0.83	0.85	4661
Default (1)	0.48	0.56	0.52	1326
accuracy			0.77	5987
macro avg	0.67	0.69	0.68	5987
weighted avg	0.78	0.77	0.77	5987

Table 3: Classification Report

6.2 Confusion Matrix

Interpretation: True Negatives (TN = 3,860) represent non-defaulting clients correctly cleared by the model. True Positives (TP = 740) are defaulters correctly flagged. The two error types are False Positives (FP = 801) — non-defaulters incorrectly labelled as at-risk — and False Negatives (FN = 586) — defaulters the model failed to detect. In credit-risk modeling, False Negatives carry the highest financial cost: a missed defaulter means the bank extends credit it will not recover. The FN count of 586 (representing 44% of all actual defaulters in the test set) is the most critical limitation of this baseline and directly motivates the use of more powerful classifiers in the Final Deliverable. False Positives (801), while generating unnecessary credit reviews, carry a lower financial risk than FNs and are a manageable trade-off when using `class_weight='balanced'`.

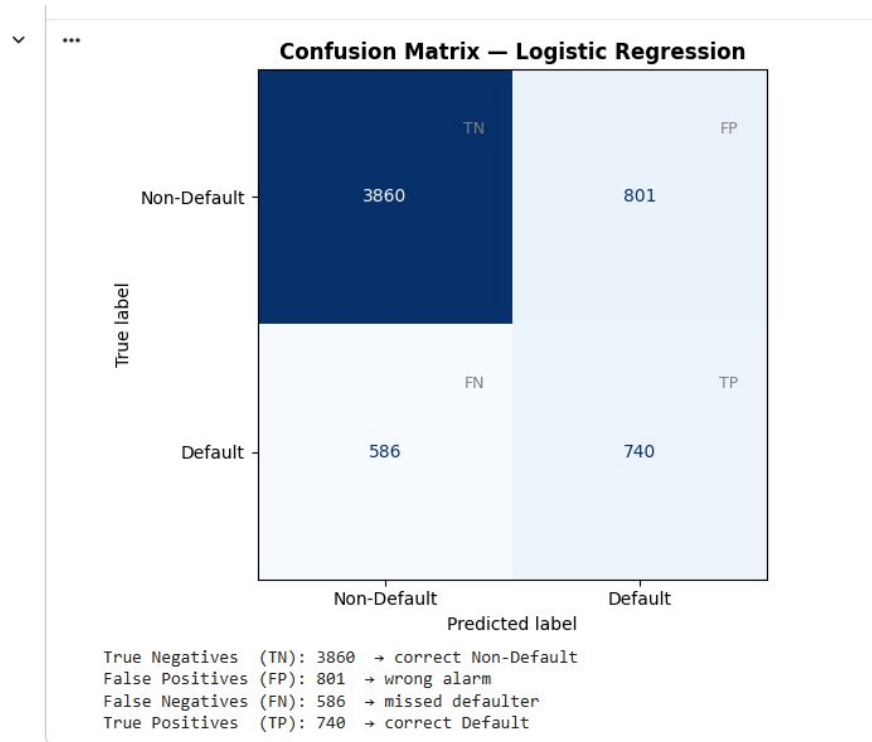


Figure 5: Confusion matrix for Logistic Regression baseline

6.3 ROC Curve

Figure 8 presents the Receiver Operating Characteristic curve for the Logistic Regression model.

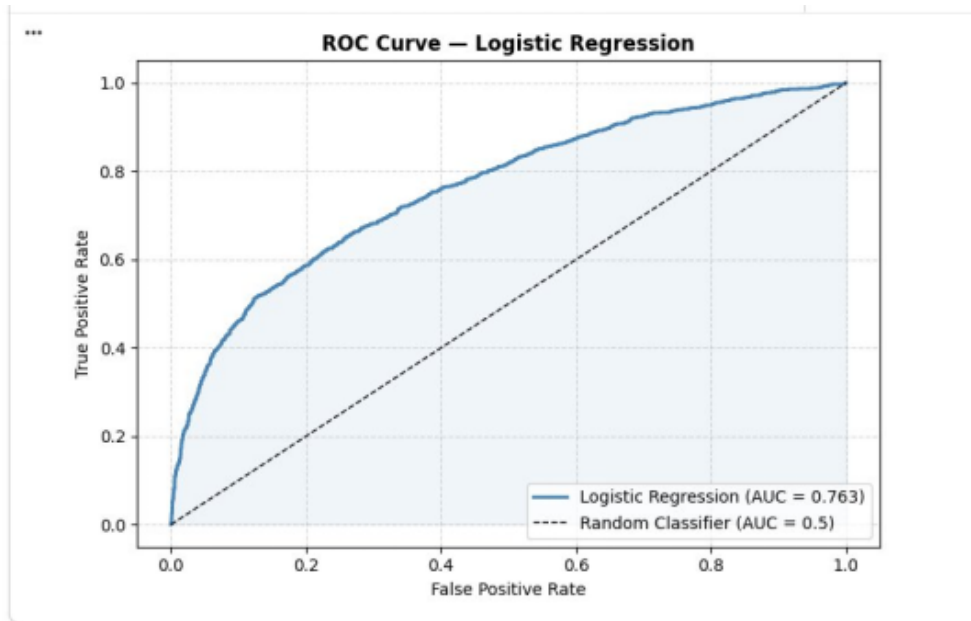


Figure 6: ROC curve of the baseline Logistic Regression model

Interpretation: The ROC curve plots the True Positive Rate (Recall) against the False Positive Rate at every classification threshold from 0 to 1. The shaded area under the curve ($AUC = 0.763$) indicates that the model correctly ranks a randomly selected defaulter above a randomly selected non-defaulter approximately 76% of the time. This is significantly above the random baseline ($AUC = 0.5$, dashed line), confirming that the feature engineering and preprocessing pipeline introduced meaningful predictive signal. The steep initial rise of the curve at low False Positive Rates indicates that the model captures a substantial fraction of defaulters with few false alarms at higher decision thresholds. This shape suggests that threshold tuning (e.g., lowering the classification threshold from 0.5 to ~ 0.3) could further improve Recall at an acceptable Precision cost — a direction to be explored in Phase 2.

6.4 Feature Coefficient Analysis

Figure 9 displays the top 15 features ranked by the absolute value of their Logistic Regression coefficients.

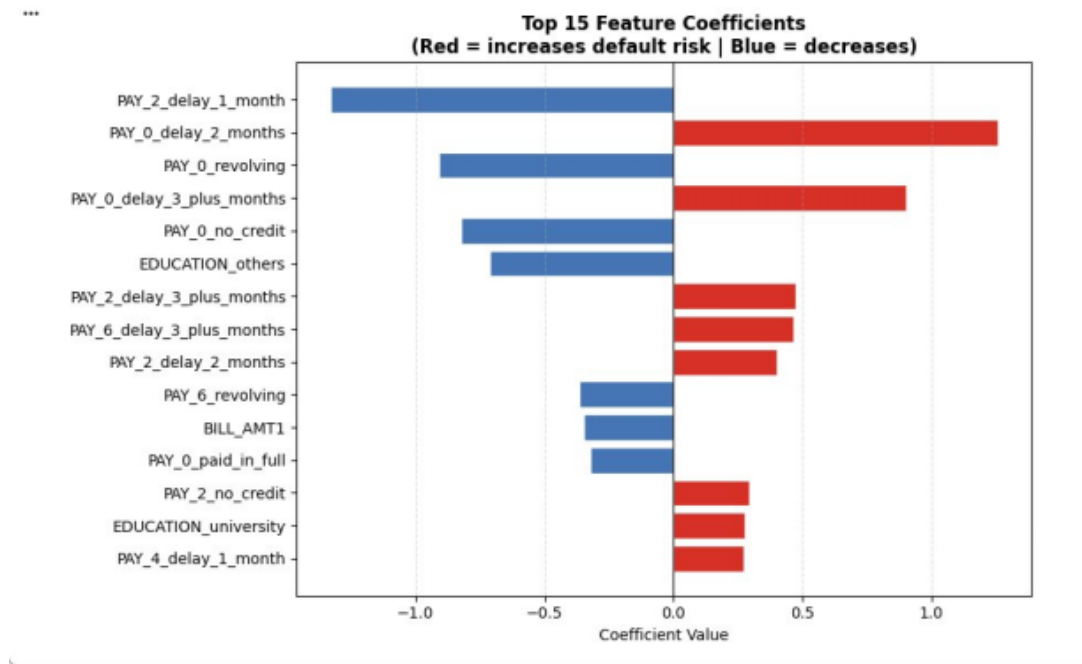


Figure 7: Top feature coefficients by absolute magnitude

Interpretation: The coefficient chart confirms the findings of the EDA analysis. `PAY_0_delay_2_months` carries the largest positive coefficient ($\approx +1.25$), making it the single strongest predictor of default — consistent with the `PAY_0` bar chart in Section 4 which showed a dramatic increase in default rate for clients with a 2-month payment delay. `PAY_0_delay_3_plus_months` ($\approx +0.90$) ranks second, confirming that recent repayment delays are the dominant risk signal in this dataset. Among negative coefficients (features that reduce default risk), `PAY_2_delay_1_month` (≈ -1.05) appears prominently — suggesting that a one-month delay in an earlier billing period is associated with lower current default risk, possibly because such clients have since resolved their payment difficulties. `EDUCATION_others` and `PAY_0_no_credit` carry moderate negative coefficients, indicating that clients with non-standard education profiles or no credit history are less likely to default in this dataset. `BILL_AMT1` and `PAY_6_revolving` show small negative effects, consistent with the expectation that higher outstanding bills and revolving balances are not the primary default driver compared to repayment behaviour.